



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO MANIPRIYA PETROL BUNK

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

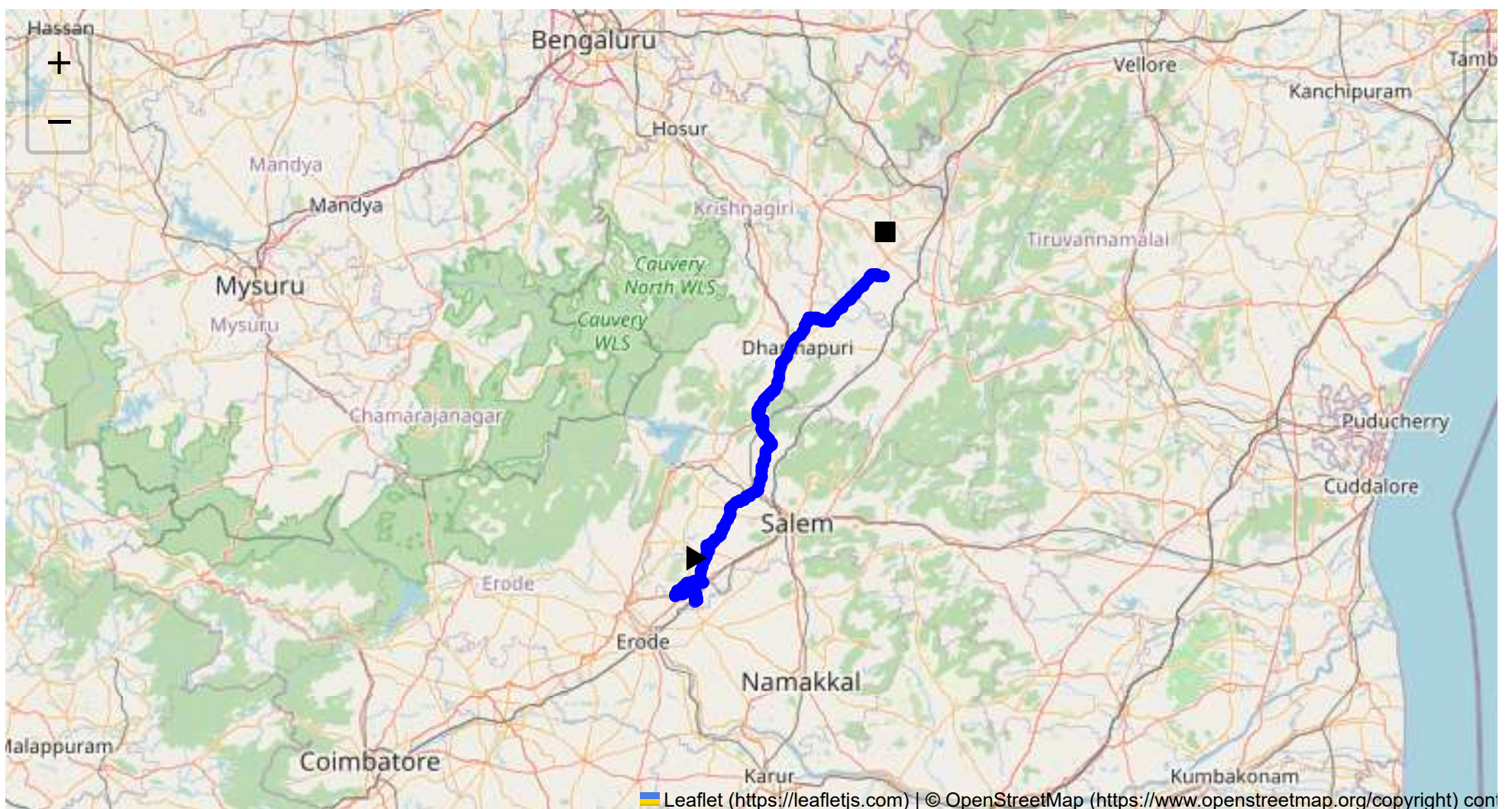
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 150.65 km
Estimated Duration: 3.3 hours
Adjusted Duration (Heavy Vehicle): 4.1 hours
Start: (11.4381, 77.8734)
End: (12.329611, 78.39942)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map: The route from Sangagiri to Balethottam spans approximately 150.65 kilometers, covering several towns and junctions across Tamil Nadu, India. Starting from Sangagiri, this route passes through key locations such as Kangeyampalayam, Padaiveedu, Tharamangalam, Omalur, Thoppur, Balajangamanhalli, Reddihalli, Manjamedu, and Melkaravoni before reaching Balethottam. This journey primarily follows major state highways and some rural roads, making it a diverse route in terms of road types.

2. Typical Weather Conditions and Potential Weather-Related Hazards: Tamil Nadu generally experiences a tropical climate. The main weather-related hazards along the route include:

- **Monsoon Rains (June to September):** Heavy rains can cause flooding and waterlogging, particularly affecting rural and less developed roads.
- **Heat Waves (March to June):** High temperatures can impact vehicle performance and driver fatigue.
- **Cyclones (October to December):** Although less frequent inland, cyclonic activities can lead to heavy rainfall and disruptions.

3. Analysis of Traffic Patterns:

- **Peak Hours:** Traffic congestion is more likely to occur during morning (7:30 AM - 10:00 AM) and evening peak hours (5:00 PM - 8:00 PM).
- **Congestion-Prone Areas:** Urban centers like Omalur and landscape bottlenecks such as narrow bridges in Padaiveedu may see higher congestion.

4. Assessment of Road Quality and Infrastructure:

- **State Highways:** Generally well-maintained, but occasional potholes and uneven surfaces can be found during the monsoon.
- **Rural Roads:** May have limited lighting, narrow passages, and varying surface quality, requiring cautious driving for heavy vehicles.

5. Suggestions for Alternative Routes for Emergencies:

- Via NH-44 for major portions to take advantage of better infrastructure in case of roadblocks on the primary route.
- Detour through major towns with adequate facilities like Salem for immediate issues.

6. Summary of Local Regulations Affecting Hazardous Material Transport:

- Require adherence to safety protocols including clear labeling, proper handling, and documentation.
- Restrictions on carrying certain types of goods through populated areas during peak times, requiring permits.

7. Overview of Historical Incidents:

- Incidents involving heavy vehicles primarily include overturning and collision due to narrow roads and high-speed areas.
- Reports of accidents increase during adverse weather conditions, stressing the need for caution.

8. Environmental Considerations and Sensitive Areas:

- Proximity to wildlife reserves near the Thoppur area, requiring noise control and speed regulations.
- Agricultural zones necessitate care to prevent contamination or spillage during transport.

9. Analysis of Communication Coverage:

- Most of the route is well-covered with telecommunications networks.
- Possible dead zones in rural stretches like the segments near Manjamedu and Reddihalli.

10. Estimated Emergency Response Times:

- Urban areas such as Omalur are equipped with emergency services potentially attending within 30 minutes.
- Rural areas might experience delays up to 1-2 hours depending on accessibility.

12. Overall Summary of Risk Assessment: The route from Sangagiri to Balethottam is moderately safe for transporting hazardous materials, given the infrastructure and environmental considerations. However, risks primarily arise from weather conditions, road quality in rural sections, and congestion during peak hours. Adherence to local transport regulations, cautious driving, and preparedness for emergencies are crucial for minimizing these risks. Regular monitoring of weather forecasts and traffic updates, along with ensuring proper vehicle maintenance, can significantly enhance the safety of the transport operation.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	U-Turn	High	11.4749755, 77.869149	10 KM/Hr
1	Turn	High	11.43968, 77.87345	15 KM/Hr
2	Turn	High	11.44028, 77.87541	15 KM/Hr
3	Turn	High	11.44898, 77.87410	15 KM/Hr
4	Turn	High	11.45007, 77.87201	15 KM/Hr
5	Turn	Medium	11.46392, 77.87328	30 KM/Hr
6	Turn	High	11.47391, 77.86844	15 KM/Hr
7	Blind Spot	Blind Spot	11.47481, 77.86920	10 KM/Hr
8	Turn	Medium	11.47354, 77.86775	30 KM/Hr
9	Turn	Medium	11.45543, 77.81780	30 KM/Hr
10	Blind Spot	Blind Spot	11.45615, 77.81677	10 KM/Hr
11	Blind Spot	Blind Spot	11.49225, 77.89560	10 KM/Hr
12	Turn	Medium	11.49377, 77.88573	30 KM/Hr
13	Turn	High	11.49469, 77.88488	15 KM/Hr
14	Turn	Medium	11.55618, 77.89789	30 KM/Hr
15	Turn	Medium	11.55699, 77.89986	30 KM/Hr
16	Turn	High	11.58228, 77.90841	15 KM/Hr
17	Turn	High	11.58340, 77.90637	15 KM/Hr
18	Turn	Medium	11.59201, 77.90649	30 KM/Hr
19	Turn	Medium	11.63882, 77.94859	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
20	Turn	High	11.64226, 77.95200	15 KM/Hr
21	Turn	Medium	11.64716, 77.95118	30 KM/Hr
22	Turn	Medium	11.69553, 77.97110	30 KM/Hr
23	Turn	Medium	11.69806, 77.97537	30 KM/Hr
24	Turn	Medium	11.71386, 77.99552	30 KM/Hr
25	Turn	Medium	11.71735, 78.00019	30 KM/Hr
26	Turn	Medium	11.71915, 78.01106	30 KM/Hr
27	Turn	Medium	11.72492, 78.01660	30 KM/Hr
28	Turn	Medium	11.72765, 78.02291	30 KM/Hr
29	Turn	Medium	11.73764, 78.03441	30 KM/Hr
30	Turn	Medium	11.73752, 78.03679	30 KM/Hr
31	Turn	High	11.73652, 78.03866	15 KM/Hr
32	Turn	Medium	11.74255, 78.04707	30 KM/Hr
33	Turn	Medium	11.77984, 78.05782	30 KM/Hr
34	Turn	Medium	11.86902, 78.08380	30 KM/Hr
35	Turn	Medium	11.87988, 78.08176	30 KM/Hr
36	Turn	Medium	11.92637, 78.06515	30 KM/Hr
37	Turn	Medium	11.93180, 78.06478	30 KM/Hr
38	Turn	Medium	11.93467, 78.06277	30 KM/Hr
39	Turn	Medium	11.93726, 78.05359	30 KM/Hr
40	Turn	Medium	11.95482, 78.04687	30 KM/Hr
41	Turn	High	11.95837, 78.04986	15 KM/Hr
42	Turn	High	11.96007, 78.04945	15 KM/Hr
43	Turn	Medium	11.96347, 78.05142	30 KM/Hr
44	Turn	High	11.96415, 78.05126	15 KM/Hr
45	Turn	Medium	11.96519, 78.04958	30 KM/Hr
46	Turn	High	11.96620, 78.04920	15 KM/Hr
47	Turn	Medium	11.96842, 78.05097	30 KM/Hr
48	Turn	Medium	12.17041, 78.16823	30 KM/Hr
49	Turn	Medium	12.21851, 78.19160	30 KM/Hr
50	Turn	Medium	12.21883, 78.19247	30 KM/Hr
51	Turn	Medium	12.21619, 78.19634	30 KM/Hr
52	Turn	Medium	12.21373, 78.21093	30 KM/Hr
53	Turn	High	12.21454, 78.21229	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
54	Turn	High	12.21200, 78.21640	15 KM/Hr
55	Turn	Medium	12.21232, 78.21749	30 KM/Hr
56	Turn	High	12.21131, 78.22316	15 KM/Hr
57	Turn	Medium	12.21082, 78.22342	30 KM/Hr
58	Turn	High	12.20520, 78.24481	15 KM/Hr
59	Turn	Medium	12.20775, 78.24745	30 KM/Hr
60	Turn	Medium	12.26519, 78.30309	30 KM/Hr
61	Turn	Medium	12.26745, 78.30399	30 KM/Hr
62	Turn	Medium	12.28438, 78.32053	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Thilagam Hospital	11.4753573, 77.8685829	30 km/h	Medium
1	clinic	Meyyazhagan Clinic	11.4752254, 77.8685671	30 km/h	Medium
2	hospital	Dr. Jaganathan Hospital	11.4752717, 77.8709312	30 km/h	Medium
3	hospital	Kathir Eye Hospital	11.4740976, 77.8681591	30 km/h	Medium
4	police	Police Quarters	11.4741243, 77.8673495	30 km/h	Medium
6	hospital	Ck Hospital	11.473504, 77.8650523	30 km/h	Medium
7	hospital	SP Hospital	11.4708567, 77.8654872	30 km/h	Medium
8	hospital	Lakshmi Hospital	11.4723746, 77.8646891	30 km/h	Medium
9	hospital	ESI Hospital	11.4717067, 77.8642668	30 km/h	Medium
10	clinic	Sutha Clinic	11.473718, 77.862614	30 km/h	Medium
12	clinic	Sun Eye Care	11.6936242, 77.9730948	30 km/h	Medium
13	hospital	Dr. Pandiyarajan Hospital	11.692704, 77.969344	30 km/h	Medium
14	hospital	Dr. Sengodan Hospital	11.6959527, 77.9695325	30 km/h	Medium
15	hospital	Omalur Government Hospital	11.74161, 78.044296	30 km/h	Medium
16	clinic	R.R. Nursing Home	11.7394129, 78.0439861	30 km/h	Medium
17	hospital	Sri Hari Hospital	11.7396726, 78.0440701	30 km/h	Medium
18	hospital	Sugumar Hospital	11.7396031, 78.0426955	30 km/h	Medium
19	hospital	Sri Sugam Hospital	11.7429821, 78.0461039	30 km/h	Medium
20	hospital	Saravana Hospital	11.7409303, 78.0458308	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
21	hospital	Arun Hospital	11.7429892, 78.0466881	30 km/h	Medium
22	clinic	Saratha Clinic	11.7422818, 78.0458175	30 km/h	Medium
24	clinic	Sri Meenakshi Nursing Home, Salem	11.7449162, 78.0474637	30 km/h	Medium
25	clinic	Divyam Child Care Clinic	11.7447135, 78.0473858	30 km/h	Medium
26	clinic	Momota Clinic	11.744982, 78.047397	30 km/h	Medium
27	clinic	Gomathi Clinic	11.7445293, 78.0470401	30 km/h	Medium
28	hospital	Vinayaka Child Hospital	11.745812, 78.0480225	30 km/h	Medium
29	hospital	Dr. G.R.Thilakam Hospital	11.746097, 78.04769	30 km/h	Medium
30	hospital	Amirthalingam Hospital	11.7456708, 78.0470548	30 km/h	Medium
31	hospital	Children Hospital	11.7458377, 78.0480205	30 km/h	Medium
32	clinic	Shri Ramana Poly Clinic	11.7477122, 78.0479293	30 km/h	Medium
33	hospital	Dr. Nagaraj Hospital	11.7464035, 78.0472401	30 km/h	Medium
34	hospital	Thulasi Hospital	11.7461516, 78.0469984	30 km/h	Medium
35	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
36	hospital	St. Joseph's Speciality Hospital	11.761611, 78.0522344	30 km/h	Medium
38	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium
40	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
41	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
42	hospital	Thoppur,Government Hospital	11.9398824, 78.0551658	30 km/h	Medium
43	hospital	Government Hospital, Pochampalli	12.3322477, 78.365767	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
5	school	Goverment Boys Higher Secondary School	11.477659, 77.8659632	30 km/h	Medium
11	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
23	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
37	school	John Britto Matriculation Higher Secondary School	11.7676907, 78.0513092	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
39	college	Dhirajlal Gandhi College of Technology	11.7968014, 78.0562095	30 km/h	Medium

Route Photos of Risky Spots

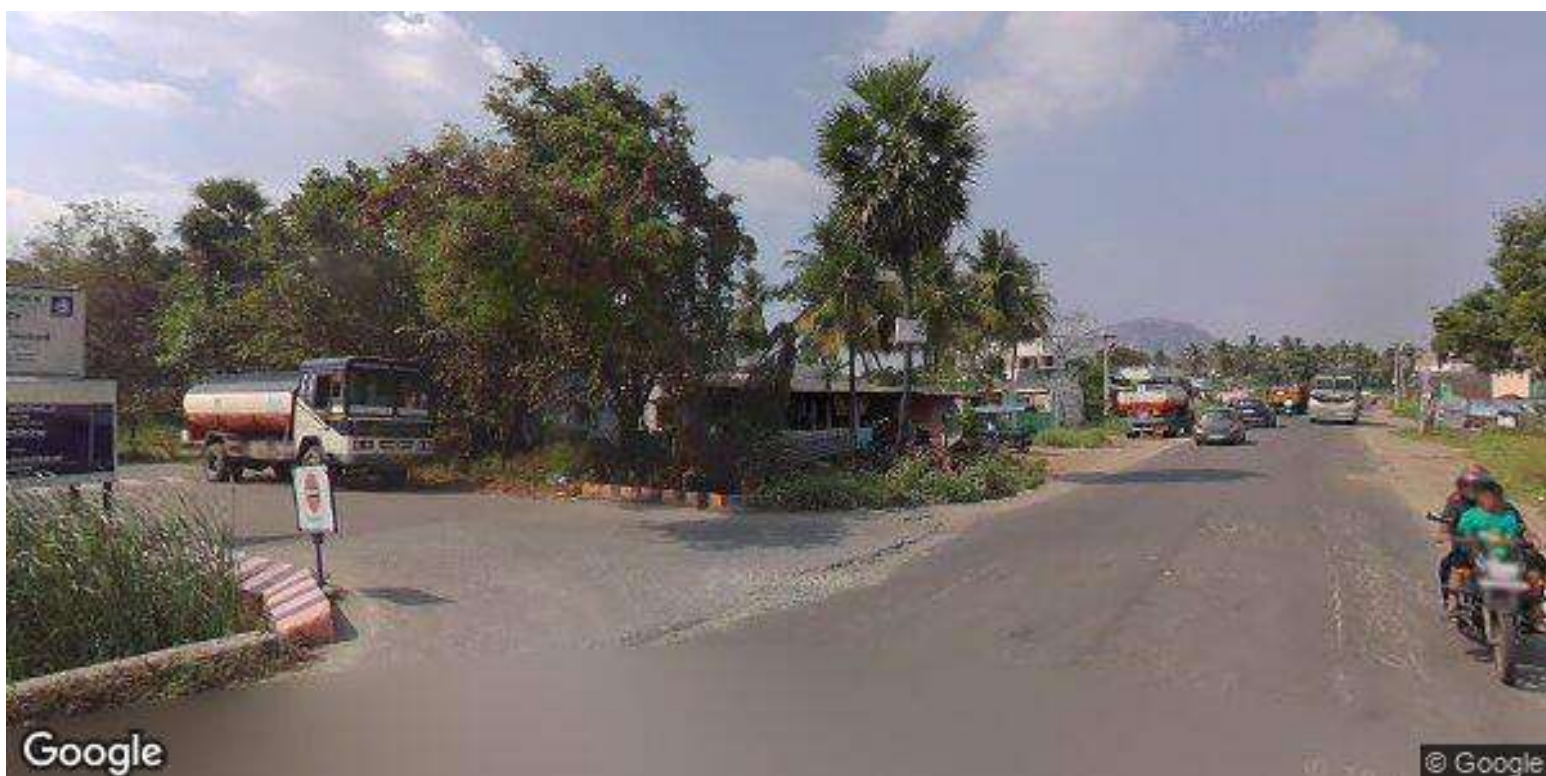


Risk Type: U-Turn

Risk Level: High

Speed Limit: 10 KM/Hr

Coordinates: 11.4749755, 77.869149



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44028, 77.87541



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44898, 77.87410



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45007, 77.87201



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46392, 77.87328



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.47391, 77.86844



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.47481, 77.86920



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.47354, 77.86775



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45543, 77.81780



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.45615, 77.81677



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.49225, 77.89560



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49377, 77.88573



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.49469, 77.88488



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.55618, 77.89789



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.58340, 77.90637



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.59201, 77.90649



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.63882, 77.94859



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.64226, 77.95200



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64716, 77.95118



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.69553, 77.97110



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.69806, 77.97537



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71386, 77.99552



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.71735, 78.00019



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.71915, 78.01106



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.72492, 78.01660



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.72765, 78.02291



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.73764, 78.03441



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.73752, 78.03679

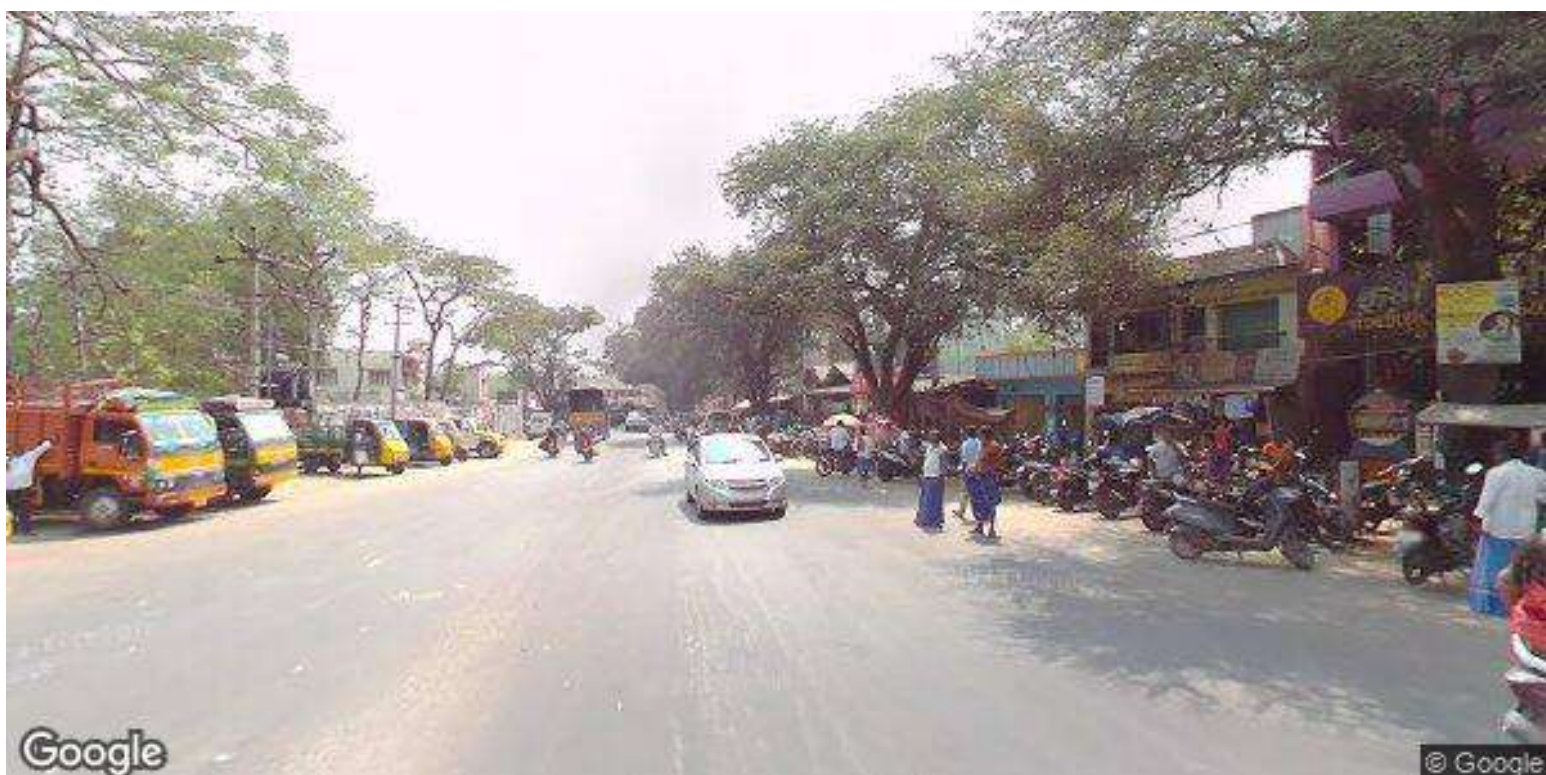


Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.73652, 78.03866



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.74255, 78.04707



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.77984, 78.05782



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.86902, 78.08380



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.87988, 78.08176



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.92637, 78.06515



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93180, 78.06478



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93467, 78.06277



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93726, 78.05359



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.95482, 78.04687



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.95837, 78.04986



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96007, 78.04945



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96347, 78.05142



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96415, 78.05126



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.96519, 78.04958



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.96620, 78.04920



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.96842, 78.05097



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.17041, 78.16823



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.21851, 78.19160



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.21883, 78.19247



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.21619, 78.19634



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.21373, 78.21093



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.21454, 78.21229



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.21200, 78.21640



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.21131, 78.22316



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.21082, 78.22342



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.20520, 78.24481



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.20775, 78.24745



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.26519, 78.30309



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.26745, 78.30399



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.28438, 78.32053
